

Q1. You are building a hospital system. Identify at least 3 classes you would create and explain what instance variables and methods each would have.

Answer:

- **Patient** → Variables: patientId, name, age. Methods: register(), updateDetails().
- **Doctor** → Variables: doctorId, name, specialization. Methods: diagnose(), prescribeMedicine().
- **Appointment** → Variables: appointmentId, date, patient, doctor. Methods: book(), cancel().

Q2. Why is it better to use private fields with getters/setters instead of making all fields public? Give a real-world scenario where this matters.

Answer:

Private fields provide data security and validation. Getters/setters control how data is accessed and modified. Example: A bank account setter can prevent a negative balance.

Q3. When would you choose a parameterized constructor over a default constructor? When would you use both together?

Answer:

Use a parameterized constructor when values are needed during object creation. Use a default constructor when values will be assigned later. Use both to give users multiple ways to create objects.

Q4. Explain in your own words: why does Java require this inside a constructor when parameter names match field names?

Answer:

this refers to the current object's instance variable. It helps Java distinguish between the field and the constructor parameter with the same name. Example: this.name = name;.

Q5. A static variable is shared across all objects. Give one scenario where this is useful, and one scenario where it could cause a bug.

Answer:

Useful: studentCount to track the total number of students.

Bug: Making name static in a Student class would cause all students to share the same name.